

Quantum Cosmology with Generalized Uncertainty Relation

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Introduction

Generalized Uncertainty Principle (GUP) and Quantum Cosmology

- The basic aim of our study is two fold,
 1. Construction of GUP Path Integral Formalism.
 2. Construction of singularity free Quantum model of Universe.
- We believe GUP has some much to do with a singularity free universe since GUP promise a minimum length scale¹.
- To show this we construct path integral expression in GUP framework and deduce eigenfunctions from it. $K(p'', p,) = \sum_n \psi_n^\dagger(p'') \psi_n(p')$
- We constructed wave packet from the eigenfunctions and choose the expansion coefficients such that the resultant wave packet follow classical-quantum correspondence². This method³ is called Canonical Initial Slope (CIS).
- Finally we study quantum singularity avoidance of the obtained wave packet.

¹P. Pedram, *Phys. Rev. D* 91, 063517 (2015).

²P. Pedram, *JCAP* 07, 006 (2008).

³S. S. Goushe *et al*, *Class. Quantum Grav.* 24, 4377 (2007).

Generalized Uncertainty Principle (GUP)

The concept of minimum length

Represented by the commutation relation⁴ $[q_i, p_j] = i\hbar\delta_{ij}[1 + \beta p^2]$; $\beta > 0$.

- The corresponding uncertainty relation is given by $\Delta q_i \Delta p_i \geq \frac{\hbar}{2}[1 + \beta(\Delta p_i)^2 + \beta p^2]$
- Offer a minimal position uncertainty $\Delta q_0 = \hbar\sqrt{\beta}$

Consequence of Generalized Uncertainty Principle

- In ordinary QM eigenstates of the operator $\mathbf{q}$ can be approximated to arbitrary precision by sequence $|\psi_n\rangle$ of physical states.
 $\lim_{n \rightarrow \infty} \Delta \mathbf{q} |\psi_n\rangle = 0$
- This is no longer the case with GUP, the existence of an absolute minimum in length implies $(\Delta \mathbf{q})^2_{|\psi\rangle} = \langle \psi | (\mathbf{q} - \langle \psi | \mathbf{q} | \psi \rangle)^2 | \psi \rangle \geq \Delta q_0$
- Though there is no position eigenstate $|q\rangle$, information on position are still accessible by realising maximally localised states $|\psi_\xi^{ml}\rangle$.

⁴ A. Kempf *et al*, *Phys. Rev.* 52, 1108 (1995)

Generalized Uncertainty Principle (GUP)

Maximally Localized State

Maximally localized state localized around a position ξ

- These states are given by the condition $\langle \psi_{\xi}^{ml} | \mathbf{q} | \psi_{\xi}^{ml} \rangle = \xi$ and $(\Delta q)_{|\psi_{\xi}^{ml}\rangle} = \Delta q_0$
- The momentum representation of these states are
$$\psi_{\xi}^{ml}(p) = \sqrt{\frac{2\sqrt{\beta}}{\pi}} (1 + \beta \mathbf{p}^2)^{-1/2} \exp\left(-i\xi \frac{\tan^{-1}(\sqrt{\beta}p)}{\hbar\sqrt{\beta}}\right)$$
- These states are completely physical and give finite energy, but in general no longer mutually orthogonal.
- To recover information on position, we project arbitrary states $|\phi\rangle$ on $|\psi_{\xi}^{ml}\rangle$ to obtain the probability amplitude of the particle being maximally localized around the position.
- We obtain the closure relation for maximally localized states as
$$(1 + \beta \mathbf{p}'^2)^{1/2} (1 + \beta \mathbf{p}^2)^{1/2} \int_{-\infty}^{\infty} \frac{d\xi}{4\hbar\sqrt{\beta}} |\psi_{\xi}^{ml}\rangle \langle \psi_{\xi}^{ml}| = 1$$
- Above closure relation together with $\int_{-\infty}^{+\infty} \frac{dp}{1+\beta p^2} |p\rangle \langle p| = 1$ are used in the construction of path integral

Generalized Uncertainty Principle (GUP)

Hamilton's Equations in GUP

- The correspondence between the QM commutator and Poisson bracket in the framework of GUP gives $\{q_i, p_j\} = (1 + \beta \mathbf{p}^2) \delta_{ij}$.
- This leads to the Hamilton's equations⁵

$$\begin{aligned}\dot{q}_i &= \{q_i, H\} = (1 + \beta \mathbf{p}^2) \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= \{p_i, H\} = -(1 + \beta \mathbf{p}^2) \frac{\partial H}{\partial q_i}\end{aligned}$$

- Weak principle of equivalence is violated in the GUP framework of classical Newtonian gravity.
- In the current case of minisuperspace model, the above equations become

$$\begin{aligned}\dot{q}^\alpha &= \{q^\alpha, H\} = (1 + \beta \mathbf{p}^2) \mathcal{N} \frac{\partial \mathcal{H}_0}{\partial p_\alpha} \\ \dot{p}_\alpha &= \{p_\alpha, H\} = -(1 + \beta \mathbf{p}^2) \mathcal{N} \frac{\partial \mathcal{H}_0}{\partial q^\alpha}\end{aligned}$$

where $\mathcal{N}$ is the Lagrange multiplier imposing the constraint $\mathcal{H}_0 = \frac{1}{2} f^{\alpha\beta}(q) p_\alpha p_\beta + V(q) = 0$. Note that the solution of these equations are not unique due to the arbitrariness of $\mathcal{N}$.

⁵ A. F. Ali. *Class. Quantum Grav.*, 28(6), 065013, (2011).

Propagator in the framework of GUP

Basic steps

- We start with the usual definition ⁶

$$\mathbf{K}(p''_{\alpha}, T; p'_{\alpha}, 0) = \lim_{N \rightarrow \infty} \langle p''_{\alpha} | e^{-i\mathcal{H}N\epsilon/\hbar} | p'_{\alpha} \rangle; \epsilon = T/N.$$
- Using the closure relations we obtained, $\mathbf{K}(p''_{\alpha}, T; p'_{\alpha}, 0) = \int \tilde{D}p_{\alpha} D\xi^{\alpha} \exp \frac{i}{\hbar} \int_0^T d\tau \left[-\xi^{\alpha} \frac{d}{d\tau} \left(\frac{\tan^{-1} \sqrt{\beta} p_{\alpha}}{\sqrt{\beta}} \right) - \mathcal{H}(\xi^{\alpha}, p_{\alpha}) \right]$, where $\int \tilde{D}p_{\alpha} = \prod_{k=1}^{N-1} \int \frac{dp_{\alpha k}}{1+\beta p_{\alpha k}^2}$ and $\int D\xi^{\alpha} = \prod_{k=1}^N \int \frac{d\xi_k^{\alpha}}{2\pi\hbar}$, note that, now Hamiltonian is the function of the new position variable ξ .
- Action $S[p_{\alpha}, \xi^{\alpha}] = \int_0^T d\tau \left[-\xi^{\alpha} \frac{d}{d\tau} \left(\frac{\tan^{-1} \sqrt{\beta} p_{\alpha}}{\sqrt{\beta}} \right) - \mathcal{H}(\xi^{\alpha}, p_{\alpha}) \right]$, when minimized give the same Hamilton's equations.
- Considering the case of minisuperspace model (Constrained system), the above action becomes

$$S[p_{\alpha}, \xi^{\alpha}, \mathcal{N},] = \int_0^T d\tau \left[-\xi^{\alpha} \frac{d}{d\tau} \left(\frac{\tan^{-1} \sqrt{\beta} p_{\alpha}}{\sqrt{\beta}} \right) - \mathcal{N} \mathcal{H}_0 \right]$$
- Non uniqueness of the solution of Hamilton's equation signals the existence of symmetries in the action.

⁶ Hoa-gang Ding et al, *Commun. Theor. Phys.*,12, 341(1989)

Propagator in the framework of GUP

Basic steps (continuation)

- It can be shown that the action is invariance under the transformation $\delta p_\alpha(\tau) = (1 + \beta p^2)\{p_\alpha, \epsilon(\tau)\mathcal{H}_0\}$, $\delta \xi^\alpha(\tau) = (1 + \beta p^2)\{\xi^\alpha, \epsilon(\tau)\mathcal{H}_0\}$
 $\delta \mathcal{N} = \dot{\epsilon}$, where $\epsilon(\tau)$ is some arbitrary function of time.
- Accordingly the variation of action is $\delta S = \left[\epsilon(\tau) \left(\xi^\alpha \frac{\partial \mathcal{H}_0}{\partial \xi^\alpha} - \mathcal{H}_0 \right) \right]_0^T$, the action vanishes if we set $\epsilon(0) = 0$ and $\epsilon(T) = 0$ and this represent reparametrization invariance.
- Invariance of the action under the reparametrization can be broken by imposing the condition⁷ $\dot{\mathcal{N}} = \chi(p_\alpha, \xi^\alpha, \mathcal{N})$.
- BRST invariant path integral expression for the propagator is obtained

$$\mathbf{K}(p''_\alpha; p'_\alpha) = \int dT \int \tilde{D}p_\alpha D\xi^\alpha \exp \frac{i}{\hbar} \int_0^T dT \left[-\xi^\alpha \frac{d}{dT} \left(\frac{\tan^{-1} \sqrt{\beta} p_\alpha}{\sqrt{\beta}} \right) - \mathcal{H}_0 \right]$$
- BRST invariance effectively dismiss the time from the dynamics which is exactly what we expect in quantum cosmology.

⁷ J. J. Halliwell, *Phys. Rev. D* 38, 2468, (1988).

Homogeneous Robertson-Walker Model

One Degree of Freedom

- Described by the metric $ds^2 = \frac{-\mathcal{N}^2(t)}{q(t)} dt^2 + q(t) d\Omega_3^2$, where $\mathcal{N}(t)$ is the lapse function, $q(t)$ is related to the scale factor and $d\Omega_3^2$ is the metric on the unit three-sphere⁸.
- Classical ADM Formalism gives the Hamiltonian in GTR. In this particular case, it is $\mathcal{H} = \frac{1}{2}(-4p^2 + \lambda\xi - 1)$, where λ is related to the cosmological constant.

- In the framework of GUP, the propagator for the above Hamiltonian is obtained to be $\mathbf{K}(p''; p') =$

$$\frac{2}{\lambda'} (1 + \beta p''^2)^{1/2} (1 + \beta p'^2)^{1/2} \exp\left\{-\frac{i}{\hbar} \frac{2}{\lambda'} \left(\frac{p''}{2} + \frac{2p'^3}{3}\right)\right\} \exp\left\{\frac{i}{\hbar} \frac{2}{\lambda'} \left(\frac{p'}{2} + \frac{2p'^3}{3}\right)\right\}$$

- We check correctness of our result by setting $\beta = 0$ and by Fourier transforming to coordinate space⁹

$$\mathbf{K}(q''; q') = \frac{4\pi^2}{(2\lambda)^{1/3}} Ai\left[\frac{1-\lambda q''}{(2\lambda)^{2/3}}\right] Ai\left[\frac{1-\lambda q'}{(2\lambda)^{2/3}}\right] \text{ and the result matches.}$$

⁸ J. J. Halliwell, *Phys. Rev. D* 38, 2468, (1988).

⁹ J. Louko, *Class. Quantum Grav.* 4, 581 (1987).

Conformally Coupled Scalar Field Quantum Cosmology

Two Degrees of Freedom

- Described by the action $S = \int d^4x \sqrt{g} \left[\frac{1}{16\pi G} \mathcal{R} - \frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi - \frac{1}{12} \mathcal{R} \phi^2 \right]$ and metric $ds^2 = -\mathcal{N}^2(t) dt^2 + a^2(t) [dr^2/(1-r^2) + r^2 d\Omega_2^2]$
- Hamiltonian becomes $\mathcal{H} = -p_a^2 + p_\chi^2 - 4(a^2 - \chi^2)$, where $\chi = a \ell_{Pl} \phi / \sqrt{2}$, $p_a = -2a\dot{a}/\mathcal{N}$ and $p_\chi = 2a\dot{\chi}/\mathcal{N}$.
- Propagator in this case become $\kappa(p_a'', p_\chi''; p_a', p_\chi') = \int dT \int \tilde{D}p_a \tilde{D}p_\chi \tilde{D}\xi_a \tilde{D}\xi_\chi$
 $\times \exp \frac{i}{\hbar} \int_0^T dT \left[-\xi_a \frac{d}{dT} \left(\frac{\tan^{-1} \sqrt{\beta} p_a}{\sqrt{\beta}} \right) - \xi_\chi \frac{d}{dT} \left(\frac{\tan^{-1} \sqrt{\beta} p_\chi}{\sqrt{\beta}} \right) + p_a^2 - p_\chi^2 + 4(\xi_a^2 - \xi_\chi^2) \right]$
- After integration we find the eigenfunctions to be

$$\psi_n(p_a, p_\chi) = 2^{2\lambda} (\Gamma(n))^2 \frac{n!(n+\lambda)\sqrt{\beta}}{2\pi\Gamma(n+2\lambda)} (1 + \beta p_a^2)^{-\lambda/2} (1 + \beta p_\chi^2)^{-\lambda/2}$$

$$\times C_n^\lambda \left(\frac{\sqrt{\beta} p_a}{\sqrt{1 + \beta p_a^2}} \right) C_n^\lambda \left(\frac{\sqrt{\beta} p_\chi}{\sqrt{1 + \beta p_\chi^2}} \right)$$
- A wave packet is constructed from the above eigenfunctions to establish a good classical-quantum correspondence¹⁰ and it is expanded as

$$\Psi(p_a, p_\chi) = \sum_{n=\text{even}} A_n \psi_n(p_a) \psi_n(p_\chi) + i \sum_{n=\text{odd}} B_n \psi_n(p_a) \psi_n(p_\chi)$$

¹⁰S. S. Goushe et al, *Class. Quantum Grav.* 24, 4377 (2007).

Conformally Coupled Scalar Field Quantum Cosmology

Two Degrees of Freedom

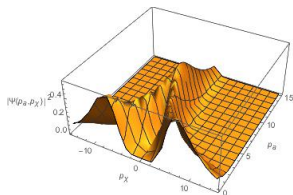


Fig.1 shows square of the wave packet obtained using the method of CIS.

The coefficient of expansion are chosen to be some distribution such that the crest of the wave packet peak around the classical path. Though the wave packet follow the classical path it is not necessary that they form a smooth function. A smooth function is obtained by adapting coherent states for the initial wave function.

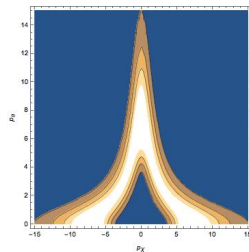


Fig. 2 shows the contour plot of fig.1 showing exactly where the wave packet peaks in the classical momentum space $p_a - p_\chi$ plane.

Finally the initial first derivative undetermined coefficients of wave packet are chosen to have the same form as initial wave function.

Future Work Plan

- As future works, the problems will be solved via the Ehrenfest theorem to observe their classical-quantum correspondence with respect to form of the potential imposing the generalised uncertainty relation.

THANK YOU